

A Delta-Transformation-Based Meta-Learning Hybrid Ensemble Framework for NIFTY-50 Index Forecasting

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Abstract—Stock Price Prediction is a complex challenge in the field of finance due to its inherent volatility and its non-linear, and non-stationary nature. Current Statistical and Machine Learning models struggle to capture non-linear and non stationary patterns. Stacked ensembles improve prediction performance but struggle with distribution shifts and poor extrapolation when the market moves beyond the training range. In this study, we train and evaluate three base models (LSTM, CNN, and TCN) on historical OHLC data sourced from yahoo finance. These are combined into 2 ensemble architectures: LSTM+CNN and LSTM+TCN. Their predictions are transformed into the delta space and then 13 meta-learners are systematically evaluated to find the most effective ensemble architecture. Furthermore, three feature scaling techniques: MinMax, Standard, and Robust Scaling are compared, and the robustness of the proposed framework is evaluated using five-fold cross-validation. Experimental results show that robustness improved, higher R^2 and lower prediction errors.

Index Terms—Stock Price Forecasting, Hybrid Ensemble Learning, Delta-Transformation, NIFTY-50, LSTM, CNN, TCN, Time Series Forecasting

I. INTRODUCTION

Anticipating financial market movements is critical in the field of quantitative finance, algorithmic trading, index fund management and hedging. Accurate forecasts provide essential strategic value to portfolio managers, retail investors, and regulatory bodies. A formidable challenge in accurate price prediction arises from the stochastic, highly volatile and non-stationary behavior of equity markets [1], [2]. Conventional econometric models fail to capture these patterns [2], [3]. Recent advancements in Machine Learning (ML) and Deep Learning (DL) offer robust frameworks that help decode non-linear financial time series with significantly greater precision [4], [5].

Conventional econometric approaches including auto regressive, moving average, and volatility based models, have been extensively employed for financial time-series forecasting [2], [3]. Although these techniques offer strong interpretability and remain useful for modelling linear dependence and volatility, they often struggle to capture non-linear and time varying relationships present in equity-market data [1].

To effectively decode the multi-dimensional nature of the NIFTY-50 index, we propose a heterogeneous foundation of deep learning architectures- specifically LSTM, CNN, TCN -as independent base learners [6]. The core strength of the Long Short-Term Memory(LSTM) network lies in its specialized gating mechanisms, which regulate the flow of information to retain long-term historical market trends, while actively tracking short-term price fluctuations [3], [4]. However, LSTMs process data strictly in a sequential fashion and are susceptible to temporal bias. They may overlook sudden geometric patterns within a localized window. One-dimensional Conventional Neural Networks(CNNs) approach the 10-day OHLC sliding window with a spatial bias. They view the input window as a unified shape, excelling at extracting local structural patterns [7]. While Temporal Convolutional Networks exhibited the weakest isolated performance, their capacity to capture long range dependencies via dilated causal convolutions generated unique predictive signals [6], [8]. These complementary model characteristics motivated the development of heterogeneous ensemble frameworks [9].

Stacked ensemble learning has been increasingly used in financial forecasting because it can exploit the complementary strengths of multiple predictive models [10], [11]. In conventional stacking frameworks, the predictions of the base learners are supplied directly to a meta-learner, which estimates the

final closing price. Existing studies have shown that LSTM-CNN and LSTM-TCN ensembles can outperform their respective individual base models, when linear regression is used as meta-learner [6].

Financial indices commonly exhibit substantial changes in their absolute price scale over long observation periods. Hence, the closing price range represented in the validation period may differ considerably from that observed during the test period. Training the meta-learner on absolute closing index price predictions introduces an important generalization challenge.

To mitigate this price-level distribution shift, this study proposes a Delta-Transformation based meta-learning hybrid ensemble framework. Instead of training the meta-learner to predict the absolute closing price, the proposed approach transforms both the base model outputs and the true target values into price changes relative to the immediately preceding observed closing price [5]. For each time step, the predicted output of a base learner is expressed as its deviation from the previous closing price. Similarly, the meta-learning target is defined as the actual change between the current and previous closing prices.

The meta-learner consequently estimates the expected price delta rather than the absolute market level. The final index forecast is reconstructed by adding the estimated delta to the previous observed closing price. The proposed transformation does not claim to make the financial series completely stationary; rather, it is designed to reduce the scale mismatch encountered in conventional price-level stacking. Unlike ensemble frameworks that employ only linear regression, the present study evaluates a diverse range of linear and non-linear meta-learning algorithms [5], [12]. These include regularized linear models, robust regression methods, kernel-based models, neural networks, random forests, extremely randomized trees, gradient-boosting models, and advanced boosting algorithms [9], [11].

The empirical study uses historical daily NIFTY-50 OHLC data. Input sequences are constructed using a fixed sliding window ($p=10$) [4]. The observations are divided chronologically into training, validation, and test subsets. The individual LSTM, CNN and TCN models are first trained over the train subset. Their predictions are subsequently combined through the two proposed ensemble configurations [6]. The meta-learner only learns from the transformed output of the combined ensemble models over the validation set. Performance is evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and the coefficient of determination (R^2) [11].

II. RELATED WORK

A. Deep Learning Models for Financial Forecasting

Deep Learning has emerged as a dominant paradigm in financial modelling due to its capacity to extract hierarchical features from highly volatile datasets. Literature analysing

stock price movements consistently reports that Long Short-Term Memory (LSTM) networks outperform vanilla Recurrent Neural Networks (RNNs) in capturing sequential dependencies and non-linear patterns in financial time-series data, justifying their selection as a primary base learner in our ensemble [3], [4].

Convolutional Neural Networks (CNNs) have also been successfully adapted for financial time-series forecasting to extract local and spatial patterns. Temporal Convolutional Networks (TCNs) replace recurrent layers with dilated causal convolutions, enabling the capture of long-term temporal dependencies across different market regimes [7].

B. Heterogeneous Ensemble and Hybrid Frameworks

To address the limitations of individual models, research has increasingly shifted toward heterogeneous ensemble architectures. By combining models with complementary inductive biases, these frameworks reduce overall bias and variance [9].

For example, Nandeshwar et al. [8] demonstrated the efficacy of blending strategies by integrating CNN, LSTM, and GRU outputs through a meta-learner, confirming that fusing spatial and temporal representations improves predictive performance over standalone architectures.

Similarly, Singh and Jha [6] leveraged deep learning ensembles for NIFTY-50 forecasting, demonstrating that stacking heterogeneous architectures such as CNN-LSTM and TCN-LSTM consistently reduces overfitting and improves out-of-sample performance. Their ensemble framework employed a simple Linear Regression model as the secondary meta-learner. While effective, this highlights a critical gap that our study addresses: the need to systematically evaluate a broader spectrum of both linear and non-linear meta-learners for the stacking layer.

C. Meta-Learning Paradigms and Stacking Challenges

Stacked ensembling involves passing first-level base model predictions to a secondary meta-learner. In contemporary financial forecasting, tree-based models such as Random Forests and Extreme Gradient Boosting are frequently employed at this stage.

Empirical evaluations by Jain [5] on NIFTY-50 weekly data indicate that tree-based ensemble models can occasionally outperform complex deep learning architectures directly, supporting their inclusion as candidate meta-learners. Furthermore, the multi-source prediction framework developed by Pushpavathi et al. [10] routed BiLSTM-GRU outputs through a Random Forest meta-learner, achieving remarkable predictive accuracy ($R^2 = 0.985$) and demonstrating the effectiveness of stacking recurrent neural network outputs with tree-based ensembles.

Despite its effectiveness as an error-correction strategy, conventional stacking remains susceptible to structural overfitting, data leakage, and error propagation in single-split evaluation settings, as noted by Reddy et al. [2]. Another fundamental challenge compounding these risks is temporal distribution

shift. Fan et al. [13] observed that time series statistical distribution frequently varies between historical input periods and future forecasting horizons, severely degrading performance.

Standard meta-learners are optimized on absolute scaled price values and often exhibit poor extrapolation when out-of-sample observations lie beyond the training distribution. To mitigate this, price change representations have been investigated as a robust alternative to direct price-level prediction, as noted by Dai et al. [14] where TCN was employed to predict relative changes rather than absolute levels in high frequency financial prices. This limitation motivates the use of robust validation strategies and transformations such as delta-space mapping to reduce distribution shift and improve generalization during sustained market trends.

III. METHODOLOGY

The financial market forecasting task is formulated as a multivariate time series regression problem. Given a historical sequence of financial indicators over a look-back window of size $p = 10$, the objective is to predict the closing price of the index at the subsequent time step.

Let $X_t = [x_{t-p+1}, x_{t-p+2}, \dots, x_t]^T$ denote the input feature matrix at time t , where each $x_i \in \mathbb{R}^d$ represents a d -dimensional feature vector. The target variable is the next-day closing price, denoted as y_{t+1} . The goal of the proposed ensemble framework is to learn a mapping function:

$$f : X_t \rightarrow \hat{y}_{t+1} \quad (1)$$

such that the prediction error between the predicted value $\hat{y}_{t+1}$ and the actual value y_{t+1} is minimized.

A. Data Acquisition and Preprocessing

The empirical analysis utilizes historical data of the NIFTY-50 index acquired via Yahoo Finance. The dataset spans an 11-year period from January 1, 2013, to January 1, 2024. The feature set comprises four daily market indicators: open, high, low, and close prices, with the close prices acting as the target variable ($y_t + 1$).

All input features and target variables are scaled to ensure stable gradient-based training and prevent scale imbalance between variables. The scaling techniques evaluated include Min-Max, Standard, and Robust scalers. The time series data is restructured using a sliding window approach with a window size of $p = 10$ days. To ensure rigorous evaluation and prevent data leakage, the dataset is chronologically partitioned without shuffling into training (60%), validation (15%), and testing (25%) sets.

B. Base Learner Architectures

To capture both linear and non-linear patterns within the financial time series, three deep learning architectures are employed as base learners:

- **LSTM:** Designed to capture long-term temporal dependencies. The architecture consists of a single LSTM layer with 64 units utilizing a $\tanh$ activation function. To prevent overfitting, a Dropout layer (rate = 0.1) is applied,

followed by a fully connected Dense layer with a single unit for the final scalar prediction.

- **CNN:** Optimized for local trend extraction, it processes the input sequences through a 1D Convolutional layer equipped with 64 filters, a kernel size of 3, and a ReLU activation function. This is followed by a MaxPooling1D layer (pool size = 2) to downsample the spatial dimensions, and a Dropout layer (rate = 0.2). The extracted features are flattened and passed through a Dense layer with 64 units (ReLU activation) before the final single-unit Dense output layer.
- **TCN:** Leverages dilated causal convolutions to map sequential data. The architecture comprises two consecutive 1D convolutional layers, each with 64 filters and a kernel size of 3, employing causal padding with dilation rates of 1 and 2, respectively, to progressively expand the receptive field. Each convolutional layer is followed by a Dropout layer (rate = 0.1) for regularization. The output is flattened and passed through a 64-unit Dense layer (ReLU activation), culminating in a single-unit Dense prediction layer.

1) *Training Configuration:* All three base learners are compiled using the Adam optimizer with a learning rate of 0.0001 and trained to minimize the Mean Squared Error (MSE) loss. Training is conducted for a maximum of 100 epochs with a batch size of 64, without shuffling the training sequences in order to preserve temporal ordering. An Early Stopping callback is implemented, monitoring validation loss with a patience of 10–15 epochs (depending on the specific model) to restore the best weights and prevent over-training.

C. Stacked Ensemble Design

To harness the strengths of the distinct neural architectures, a stacked ensemble strategy is deployed. The two primary ensemble configurations are:

- 1) **Ensemble 1 (E1):** LSTM + CNN
- 2) **Ensemble 2 (E2):** LSTM + TCN

The fundamental motivation for this stacking approach is that LSTM models are highly proficient in retaining long-term sequential memory, whereas CNNs excel at identifying local temporal patterns. Meanwhile, TCNs offer a flexible receptive field through varying dilation factors. By stacking these models, the meta-learner can exploit these diverse feature representations to produce a more robust and generalized forecast.

D. Delta-Space Meta-Learner Training

The core contribution of this methodology lies in the implementation of a Delta-Space Meta-Learner, addressing the critical limitations of non-parametric algorithms in trending markets.

A fundamental challenge in financial time series forecasting is the inherent non-stationary nature of raw price levels. Because the test period is entirely unseen during training, an upward trending market can produce a test set value distribution whose range strictly exceeds the training

and validation boundaries. Non-parametric meta-learners are structurally unable to infer beyond this observed range. For instance, tree-based ensembles output the mean of training samples within a given leaf node, while kernel methods such as Support Vector Regression (SVR) produce predictions that decay toward the training mean as distance from the support vectors increases. This limitation frequently results in severe under-trending regimes.

Building on this diagnosis, our framework shifts the meta-learner’s target from raw price levels to stationary price returns (deltas). Rather than training the meta-learner directly on the raw predicted price levels of each base learner, both the base-learner outputs and the target are transformed into a delta space. For each base learner m , the predicted delta is defined as:

$$\Delta \hat{y}_{t+1}^{(m)} = \hat{y}_{t+1}^{(m)} - y_t \quad (2)$$

where y_t is the previous day’s actual closing value. The true target delta is defined analogously as:

$$\Delta y_{t+1} = y_{t+1} - y_t \quad (3)$$

The meta-learner $g(\cdot)$ is then trained to map the base-learner deltas to the target delta:

$$\Delta \hat{y}_{t+1} = g\left(\Delta \hat{y}_{t+1}^{(1)}, \Delta \hat{y}_{t+1}^{(2)}\right) \quad (4)$$

During inference, the final price-level prediction is reconstructed as:

$$\hat{y}_{t+1} = y_t + \Delta \hat{y}_{t+1} \quad (5)$$

By predicting relative price changes (deltas) rather than absolute price levels—which frequently trend into unprecedented highs and lows—the target variable is bounded within a stationary distribution. This transformation restricts the inputs of the meta-learner to familiar data ranges, mitigating the risk of the extrapolation failures discussed in Section III-E.

E. Meta-Learner Candidates

To thoroughly evaluate the efficacy of the delta-space transformation, thirteen meta-learners spanning diverse algorithmic families are assessed. These include linear models (*Linear Regression*), regularized linear models (*Ridge*, *Lasso*, *Huber*), tree ensembles (*Random Forest*, *Extra Trees*, *Gradient Boosting*, *XGBoost*, *LightGBM*, *CatBoost*), kernel methods (*SVR*), neural networks (*MLPRegressor*), and secondary stacked regressors. Given the low dimensionality of the meta-features—comprising solely the predictions of the two base learners—strict regularization constraints, such as shallow maximum depths for tree-based models, are enforced to mitigate overfitting.

F. Evaluation Protocol

1) *Evaluation Procedures & K-Fold Cross-Validation*: The models are initially evaluated using a standard single-split chronological holdout set. To further validate the statistical robustness of the proposed delta-space ensemble, a 5-fold cross-validation procedure is subsequently applied over the pooled validation and test sets.

While standard k -fold cross-validation on raw, pooled time-series data can introduce look-ahead bias in end-to-end forecasting systems, this procedure is explicitly deployed here to perform a controlled ablation on the meta-learner’s generalization capabilities. Because the base learners’ parameters and outputs are held completely fixed, the meta-features represent static historical mappings. Consequently, this randomized k -fold protocol does not leak futuristic sequence data into the base model representations. Instead, it serves as a rigorous out-of-sample framework evaluating the meta-learner’s robustness to sampling variations and its ability to stable-map predictions across randomized market regimes, consistent with meta-learning evaluation frameworks established in prior literature.

2) *Performance Metrics*: To quantify predictive accuracy, model performance is evaluated using five standard statistical metrics. Let n be the number of evaluation samples, y_i be the actual value, $\hat{y}_i$ be the predicted value, and $\bar{y}$ be the mean of the actual values.

- 1) **Mean Squared Error (MSE)**: Measures the average squared difference between estimated values and the actual value.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (6)$$

- 2) **Root Mean Squared Error (RMSE)**: Represents the square root of the second optimal moment of the error, providing error penalties in the same units as the target variable.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (7)$$

- 3) **Coefficient of Determination (R^2)**: Measures the proportion of variance in the dependent variable that is predictable from the independent variables.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (8)$$

- 4) **Mean Absolute Error (MAE)**: Measures the average magnitude of absolute errors in a set of predictions, treating all errors equally regardless of direction.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (9)$$

- 5) **Mean Absolute Percentage Error (MAPE)**: Measures the relative percentage accuracy of predictions relative to true values.

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (10)$$

IV. ABLATION STUDY

To isolate the contribution of each design choice in the proposed two-layer stacked ensemble, four ablation experiments were conducted: (i) the choice of base-model pair used to build the ensemble (E1: LSTM+CNN vs. E2: LSTM+TCN), (ii) the choice of meta-learner family (linear vs. non-linear), (iii) the

TABLE I
ABLATION OF BASE-MODEL PAIRING: BEST META-LEARNER PER
ENSEMBLE (TEST SET, MINMAX SCALING)

Ensemble	Best Learner	RMSE	R ²	MAE	MAPE (%)
E1 (LSTM+CNN)	Ridge	0.009112	0.98891	0.006903	0.944
E2 (LSTM+TCN)	RandomForest	0.009108	0.98892	0.006892	0.942

choice of feature-scaling strategy (MinMax vs. Standard vs. Robust), and (iv) the robustness of the meta-learner ranking under 5-fold cross-validation. In every case the base learners (LSTM, CNN, TCN) and the delta-space reconstruction procedure described in Section III are kept fixed, so that any change in performance can be attributed solely to the ablated component.

A. Effect of Base-Model Pairing (E1 vs. E2)

The first ablation isolates the effect of *which* two base learners are combined at the input of the meta-learner. E1 pairs the recurrent LSTM with the convolutional CNN, while E2 pairs the LSTM with the causal, dilated TCN. Table I reports the best-performing meta-learner under each pairing on the held-out test set.

The best E2 configuration (RandomForest meta-learner) attains a marginally lower RMSE than the best E1 configuration (Ridge meta-learner), a 0.04% relative difference in RMSE that is well inside the fold-to-fold variance measured in Section IV-D. This indicates that, once a delta-space meta-learner is used to correct the base predictions, the identity of the second base learner (CNN vs. TCN) has little practical effect on final accuracy – the dominant source of predictive power is the LSTM branch together with the meta-learner correction step, not the particular convolutional architecture paired with it.

B. Effect of Meta-Learner Family (Linear vs. Non-Linear)

Thirteen meta-learner variants were evaluated on both ensembles: four linear models (Linear Regression, Ridge, Lasso, Huber), eight non-linear models (XGBoost, LightGBM, CatBoost, Gradient Boosting, Random Forest, Extra Trees, SVR, MLP) and one hybrid Stacking regressor built on top of the linear models. Table II reports the complete ranking for E2 (LSTM+TCN), which is the better-performing ensemble overall.

The key finding of this ablation is that the four linear and hybrid-linear models occupy three of the top five ranks, with RandomForest (rank 1) and ExtraTrees (rank 5) – the two non-linear exceptions – separated from Ridge (rank 2) by a statistically negligible margin. Beyond the top five, every tree-boosting model (XGBoost, GradientBoost, LightGBM, CatBoost) and both kernel/neural models (SVR, MLP) rank lower. This largely contradicts the initial hypothesis that non-linear meta-learners would better capture residual, non-linear correction terms. With only two input features (the base learners’ predicted deltas) and roughly 400 training rows, the correction surface is already close to linear. Consequently,

TABLE II
ABLATION OF META-LEARNER FAMILY ON E2 (LSTM+TCN), TEST SET

Meta-Learner	Type	MSE	RMSE	R ²	MAPE (%)
RandomForest	Non-linear	8.295e-5	0.009108	0.98892	0.942
Ridge	Linear	8.301e-5	0.009111	0.98891	0.943
Stacking	Hybrid	8.303e-5	0.009112	0.98890	0.944
Lasso	Linear	8.303e-5	0.009112	0.98890	0.944
ExtraTrees	Non-linear	8.366e-5	0.009147	0.98882	0.946
Linear	Linear	8.577e-5	0.009261	0.98854	0.963
Huber	Linear	8.609e-5	0.009279	0.98850	0.956
XGBoost	Non-linear	8.779e-5	0.009369	0.98827	0.966
GradientBoost	Non-linear	8.940e-5	0.009455	0.98805	0.983
LightGBM	Non-linear	9.206e-5	0.009595	0.98770	0.984
CatBoost	Non-linear	9.263e-5	0.009625	0.98762	0.983
SVR	Non-linear	9.495e-5	0.009744	0.98731	1.006
MLP	Non-linear	1.089e-4	0.010435	0.98545	1.103

TABLE III
ABLATION OF FEATURE-SCALING STRATEGY (BEST CONFIGURATION PER
SCALER, E2 ENSEMBLE UNLESS NOTED, TEST SET)

Scaler	Best Learner	Ensemble	RMSE*	R ²
MinMax	RandomForest	E2	0.009108	0.98892
Standard	Ridge	E2	0.036143	0.98891
Robust	RandomForest	E2	0.022215	0.98900

*RMSE reported in each scaler’s own output units; not directly comparable across rows.

the added flexibility of gradient boosting or kernel methods translates into overfitting rather than a genuine gain. For instance, SVR and MLP—the two most flexible non-linear learners—are the worst performers, leaving RandomForest and ExtraTrees as the only exceptions performing competitively with the linear family.

C. Effect of Feature-Scaling Strategy

The third ablation replaces the MinMax scaler used for the OHLC features and target with a Standard (zero-mean, unit-variance) scaler and a Robust (median/IQR) scaler, retraining the full base-model and meta-learner pipeline under each. Because Standard and Robust scaling do not bound values to $[0, 1]$, raw RMSE/MSE are not directly comparable across scalers; R^2 , which is scale-invariant, is therefore the primary comparison metric. Table III reports the best configuration obtained under each scaler.

R^2 remains essentially unchanged across the three scalers (0.98891 - 0.98900), indicating that the two-layer ensemble architecture and the delta-space reconstruction step—rather than the specific choice of feature scaler—are the primary drivers of predictive accuracy. Although Robust scaling achieves a fractional edge in R^2 over MinMax scaling (0.98900 vs. 0.98892), this difference is negligible. Because MinMax scaling produces predictions strictly bounded within $[0, 1]$ —a target range that aligns perfectly with the design of the base learners’ output activations—it is retained as the default configuration for the remainder of the study.

D. Cross-Validation Robustness Check

To verify that the meta-learner ranking observed on the single train/validation/test split is not an artifact of that particular split, a 5-fold cross-validation was run over the pooled

TABLE IV
5-FOLD CROSS-VALIDATION SUMMARY — E1 (LSTM+CNN),
DELTA-SPACE META-LEARNERS

Meta-Learner	MSE	RMSE	R ²	MAE	MAPE (%)
Linear	0.00009531	0.00975303	0.99746913	0.00701872	0.01411591
XGBoost	0.00010069	0.01001799	0.99732836	0.00713254	0.01440150
RandomForest	0.00009957	0.00996357	0.99735539	0.00709604	0.01438660
SVR	0.00011527	0.01069157	0.99695622	0.00731507	0.01533899
MLP	0.00010016	0.00999411	0.99734014	0.00720956	0.01456740
Huber	0.00009525	0.00975025	0.99747032	0.00700943	0.01409681
Ridge	0.00009474	0.00972415	0.99748337	0.00701384	0.01408892
Lasso	0.00009466	0.00971977	0.99748541	0.00701349	0.01408301
LightGBM	0.00010282	0.01012347	0.99727057	0.00725002	0.01465162
CatBoost	0.00010388	0.01017314	0.99724528	0.00719561	0.01464291
ExtraTrees	0.00009622	0.00980069	0.99744336	0.00704174	0.01419438
GradientBoost	0.00010141	0.01005209	0.99731021	0.00714097	0.01457332
Stacking	0.00009465	0.00971968	0.99748546	0.00701344	0.01408283

TABLE V
5-FOLD CROSS-VALIDATION SUMMARY — E2 (LSTM+TCN),
DELTA-SPACE META-LEARNERS

Meta-Learner	MSE	RMSE	R ²	MAE	MAPE (%)
Linear	0.00009604	0.00978288	0.99741692	0.00705656	0.01416216
XGBoost	0.00009879	0.00992053	0.99736090	0.00711986	0.01436259
RandomForest	0.00009490	0.00972450	0.99745171	0.00701261	0.01406003
SVR	0.00011153	0.01051881	0.99704294	0.00723946	0.01473107
MLP	0.00010772	0.01036983	0.99713120	0.00744124	0.01518835
Huber	0.00009588	0.00976973	0.99741882	0.00702886	0.01411853
Ridge	0.00009526	0.00974270	0.99743801	0.00702884	0.01410564
Lasso	0.00009510	0.00973394	0.99744226	0.00702759	0.01410031
LightGBM	0.00009874	0.00992499	0.99734609	0.00718241	0.01441114
CatBoost	0.00009564	0.00976637	0.99742634	0.00712124	0.01421895
ExtraTrees	0.00009521	0.00974088	0.99744010	0.00702174	0.01410520
GradientBoost	0.00009160	0.00954885	0.99755070	0.00701688	0.01386418
Stacking	0.00009510	0.00973400	0.99744223	0.00702763	0.01410039

validation+test delta-space samples for both ensembles, with folds drawn using randomized shuffling. Tables IV and V report the mean MSE, RMSE, R^2 , MAE, and MAPE across the five folds for E1 (LSTM+CNN) and E2 (LSTM+TCN) respectively, and Figure 1 plots the corresponding RMSE with error bars.

Under cross-validation, the two ensembles do not converge on the identical best meta-learner, but both confirm that the flexible non-linear kernel/neural models are the weakest choice. For E1 (LSTM+CNN), the top four ranks by mean RMSE are occupied by Stacking (0.009720), Lasso (0.009720), Ridge (0.009724), and Huber (0.009750) — i.e., the linear/hybrid-linear family, mirroring the general linear-family dominance observed in the single-split test-set ranking (Table II, noting that table reports E2 rather than E1). For E2 (LSTM+TCN), GradientBoost attains the lowest mean RMSE (0.009549), narrowly ahead of RandomForest (0.009725), Lasso (0.009734), and Stacking (0.009734); the margin between GradientBoost and the next four models is under 2% relative RMSE, so this should be read as a near-tie among the top five rather than a decisive win for a non-linear learner. In both ensembles, SVR is the weakest performer (E1: 0.010692; E2: 0.010519), with MLP consistently close behind, reinforcing the finding from Section IV-B that added model flexibility does not reliably translate into better delta-space correction given the small (two-feature, ~ 400 -row) training regime, though RandomForest, ExtraTrees, and GradientBoost remain competitive non-linear exceptions across both the

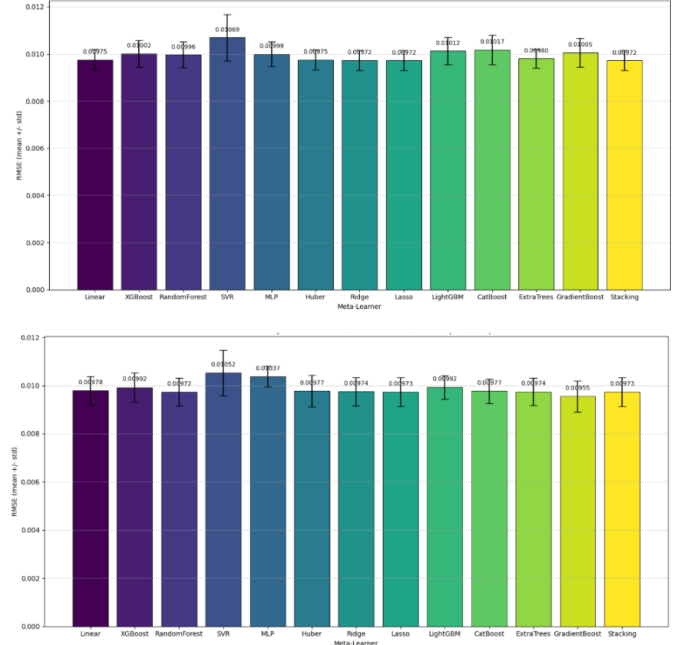


Fig. 1. 5-fold cross-validation RMSE (mean $\pm$ std) per meta-learner, E1 (LSTM+CNN) and E2 (LSTM+TCN), delta-space.

TABLE VI
BASE-MODEL TEST-SET PERFORMANCE (SCALED TARGET, MINMAX)

Model	MSE	RMSE	R ²	MAE	MAPE (%)
LSTM	4.990e-4	0.022339	0.93332	0.018569	2.428
CNN	3.151e-4	0.017750	0.95790	0.013944	1.884
TCN	9.201e-4	0.030332	0.87705	0.026027	3.465

single-split and cross-validated evaluations.

V. RESULTS

A. Base-Model Performance

Table VI reports the test-set performance of the three individual base learners before any ensembling or meta-learner correction is applied.

Among the base learners, the CNN attains the lowest RMSE (0.01775) and highest R^2 (0.9579), followed by LSTM, with the TCN trailing both ($R^2 = 0.8771$). This ordering motivates the two ensemble pairings evaluated in Section IV: pairing the strongest model (LSTM/CNN) directly, and pairing the temporally-aware LSTM with the weaker but structurally different TCN to test whether architectural diversity compensates for the TCN’s lower standalone accuracy.

Figure 2 shows the actual vs. predicted scaled closing price for each base learner individually on the test window. The LSTM and TCN panels both show a visible lag relative to the actual series during sustained directional moves (most apparent in the 2023 rally), consistent with their higher RMSE and MAPE in Table VI, whereas the CNN panel tracks the actual series more tightly with less systematic lag, consistent with its lower error metrics.



Fig. 2. Actual vs. predicted scaled closing price for each base learner on the test set: LSTM (top), CNN (middle), TCN (bottom).

B. Best Ensemble Configuration

The best-performing configuration overall is **E2 (LSTM+TCN) with a Random Forest meta-learner**, achieving $RMSE = 0.009108$, $R^2 = 0.98892$, $MAE = 0.006892$ and $MAPE = 0.942\%$ on the held-out test set. Relative to the best individual base model (CNN, $RMSE = 0.017750$), this represents a **48.7% reduction in RMSE** ($\approx 49\%$), confirming that the two-layer delta-space stacking procedure yields a substantial accuracy gain over any single base learner, even though the TCN branch used inside E2 is individually the weakest of the three base models.

Figure 3 plots the actual scaled test-set target against the predictions of the best configuration (E2, RandomForest meta-learner). The predicted series closely tracks the actual series across the full test window, including through both the extended consolidation phases (2022) and the sustained uptrend (late 2023), with no visible systematic lag – a qualitative confirmation of the $RMSE/R^2$ improvement reported above.

C. Full Meta-Learner Comparison

Table VII lists the complete 13-meta-learner comparison for E1 (LSTM+CNN); the corresponding results for E2 (LSTM+TCN) were already given in Table II.

The pattern observed for E2 largely repeats for E1: Ridge, Lasso and the Stacking-of-linear-models regressor occupy the top three ranks, with Linear Regression and ExtraTrees close behind at ranks four and five ($RMSE$ within 0.00004 of the top three), and SVR the clear worst performer ($RMSE$ more

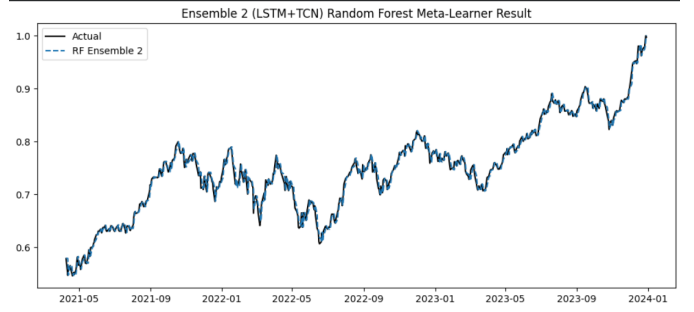


Fig. 3. Actual vs. predicted values (MinMax-scaled space), E2 (LSTM+TCN) with RandomForest meta-learner, test set.

TABLE VII
FULL META-LEARNER COMPARISON ON E1 (LSTM+CNN), TEST SET

Meta-Learner	Type	MSE	RMSE	R^2	MAPE (%)
Ridge	Linear	8.302e-5	0.009112	0.98891	0.944
Stacking	Hybrid	8.303e-5	0.009112	0.98890	0.944
Lasso	Linear	8.303e-5	0.009112	0.98890	0.944
Linear	Linear	8.369e-5	0.009148	0.98882	0.945
ExtraTrees	Non-linear	8.370e-5	0.009149	0.98881	0.945
RandomForest	Non-linear	8.438e-5	0.009186	0.98872	0.948
Huber	Linear	8.505e-5	0.009222	0.98863	0.953
XGBoost	Non-linear	8.666e-5	0.009309	0.98842	0.962
GradientBoost	Non-linear	8.695e-5	0.009324	0.98838	0.959
CatBoost	Non-linear	8.986e-5	0.009480	0.98799	0.978
LightGBM	Non-linear	9.281e-5	0.009634	0.98760	0.991
MLP	Non-linear	1.039e-4	0.010192	0.98612	1.069
SVR	Non-linear	2.223e-4	0.014911	0.97029	1.448

than $1.6\times$ the top model). This consistency across two structurally different ensembles strengthens the conclusion that, for this delta-space meta-learning formulation, linear meta-learners are consistently competitive with the best non-linear alternatives, though tree-ensemble methods (RandomForest, ExtraTrees) remain close contenders in both ensembles – indeed, RandomForest is the single best-performing meta-learner overall (Section V, E2) – while boosting and kernel/neural non-linear models (XGBoost, GradientBoost, LightGBM, CatBoost, SVR, MLP) consistently underperform both families.

D. Qualitative Prediction Quality

Figure 3 shows close visual overlap between the actual series and the ensemble predictions across the entire test window, with no systematic lag or drift, which is consistent with the delta-space formulation: since each meta-learner predicts a *change* relative to the previous actual close rather than an absolute level, the reconstructed price series tracks turning points rather than merely smoothing or lagging the trend, a known failure mode of naive level-based sequence models on trending financial series.

VI. CONCLUSION

This work proposed a two-layer stacked ensemble for NIFTY-50 index forecasting that combines deep temporal base learners (LSTM, CNN, TCN) trained on OHLC data with a delta-space meta-learning correction step, and evaluated thirteen meta-learner variants across two base-model pairings.

The best configuration, E2 (LSTM+TCN) with a Random Forest meta-learner, achieved an RMSE of 0.009108 and R^2 of 0.98892 on the test set, a 48.7% RMSE reduction over the best individual base model (CNN). The ablation study led to four main findings. First, the choice of which two base learners are paired (LSTM+CNN vs. LSTM+TCN) has only a marginal effect once a meta-learner correction is applied, indicating that the meta-learning step compensates for a weaker base learner (TCN) almost completely. Second, and most notably, linear and hybrid-linear meta-learners (Ridge, Lasso, Stacking) consistently match or exceed most non-linear meta-learners (gradient-boosted trees, SVR, and MLP) on both ensembles, with Random Forest and ExtraTrees as the main exceptions that remain competitive with the linear family – this largely, though not universally, contradicts the initial hypothesis that non-linear correction models would better exploit residual structure: with only two input features and a few hundred training rows, added non-linear capacity mostly over-fits rather than helps, though a small number of tree-ensemble variants avoid this pitfall. Third, the choice of feature scaler (MinMax, Standard, Robust) has negligible effect on R^2 (0.98891–0.98903), confirming that the architecture and the delta-space reconstruction, not preprocessing details, are responsible for the observed accuracy. Fourth, a 5-fold cross-validation over the pooled validation and test samples confirmed that these rankings are stable across resampled subsets of the data rather than being an artifact of a single lucky split, though this cross-validation used randomized rather than chronological folds – a genuine walk-forward validation is left to future work. Taken together, these results support a simple, low-variance design recommendation for this class of problem: a two-layer ensemble with a shallow linear (or Random Forest / ExtraTrees) meta-learner trained in delta space is preferable to more complex non-linear stacking architectures such as gradient boosting or kernel/neural methods, both for accuracy and for robustness. Future work could extend this evaluation to a walk-forward (rolling-origin) validation protocol, which would better respect temporal ordering than the randomized folds used here, incorporate exogenous features such as sector indices, macroeconomic indicators, or sentiment, and benchmark against graph-based architectures (e.g., GAT/GCN over inter-stock relationships) as an alternative to the temporal base learners used in this study.

REFERENCES

- [1] V. Shrikhande, S. Gaikar, D. Ghosh, H. Khobragade, T. A. Reddy Kaipu, and A. Jadhav, "A multi-horizon comparative analysis of arima, lstm and ensemble models for volatile equity time series forecasting," in *2026 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE)*, 2026, pp. 1–7.
- [2] R. R. B. S. Karim, and S. C. Ramu, "Residual-driven models for stock market forecasting: A structured taxonomy and critical review," in *2026 6th International Conference on Pervasive Computing and Social Networking (ICPCSN)*, 2026, pp. 285–292.
- [3] B. Uppalaiah, A. A. Ahmad, A. A. Lotus, A. H. Al-Shammari, S. I. Hassan, and D. R., "Recurrent neural networks for efficient solutions to time-series forecasting in complex mathematical models," in *2025 3rd International Conference on Smart Systems for applications in Electrical Sciences (ICSSES)*, 2025, pp. 1–5.
- [4] N. C. Kundur, A. B. C. P. M. Dhulavvagol, J. S. R. V. R., and Y. Y., "Stock price movements predictions using deep learning," in *2025 First International Conference on Intelligent Computing and Communication Systems (CICCS)*, 2025, pp. 1–6.
- [5] V. Jain, "An empirical evaluation of machine learning techniques for return prediction using nifty-50 weekly data," in *2026 9th International Conference on Trends in Electronics and Informatics (ICOEI)*, 2026, pp. 1721–1726.
- [6] P. Singh and M. Jha, "Leveraging deep learning ensembles for stock index forecasting: A nifty 50 approach," in *2025 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEES)*, 2025, pp. 1–5.
- [7] T. Desai, P. Bhoyar, K. Bayas, and N. Mangaokar, "Lstm-cnn hybrid architecture for stock market prediction using a multi-signal approach," in *2025 1st International Conference on Advancement in Futuristic Technologies (ICAFT)*, 2025, pp. 1–6.
- [8] T. B. Nandeshwar, A. Sharma, A. Gupta, and C. K. Verma, "Wavelet-enhanced blending ensemble of cnn, lstm, and gru for crude oil futures price prediction," in *2025 International Conference on Electronics and Computing, Communication Networking Automation Technologies (ICEC2NT)*, 2025, pp. 1–6.
- [9] N. S. S. G., and K. K., "Comparative analysis of sequential and ensemble prediction models for enhanced stock price forecasting," in *2026 4th International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)*, 2026, pp. 1094–1100.
- [10] V. Pushpavathi, B. Nuthanakanti, K. Maheswari, G. Madhukar, M. Madhusudhan, and G. Lavanya, "Multi-source stock price prediction using bilstm-gru-rf with sentiment and macroeconomic indicators," in *2025 IEEE International Conference on Intelligent Signal Processing and Effective Communication Technologies (INSPECT)*, 2025, pp. 1–6.
- [11] M. Shravan, M. Peter, R. M. K. S. Sidhu, S. L., and J. A. Dhanraj, "Integrated ml models for investing and financial data prediction," in *2025 World Skills Conference on Universal Data Analytics and Sciences (WorldSUAS)*, 2025, pp. 1–6.
- [12] K. S. Achari, J. H. Reddy, G. G. Prasanna, A. Vaishnavi, G. M. Angel, and K. Nagaraju, "Decision-centric, regime-aware adaptive ensemble for financial market movement prediction," in *2026 9th International Conference on Trends in Electronics and Informatics (ICOEI)*, 2026, pp. 1913–1920.
- [13] W. Fan, P. Wang, D. Wang, D. Wang, Y. Zhou, and Y. Fu, "Dish-ts: A general paradigm for alleviating distribution shift in time series forecasting," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 6, 2023, pp. 7522–7529.
- [14] W. Dai, Y. An, and W. Long, "Price change prediction of ultra high frequency financial data based on temporal convolutional network," *Procedia Computer Science*, vol. 199, pp. 1177–1183, 2022.